

A Resource Theory of Shared Complex Structure in Real Quantum Composition

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Whether real Hilbert spaces can support quantum theory is a question that lives in composition, not in individual systems. A single complex system admits a real reformulation by replacing multiplication by i with an orthogonal complex structure J . When two such systems are combined, however, the full real tensor product is strictly larger than the realification of the complex tensor product, and the extra degrees of freedom encode disagreement between the two local complex structures. We show that ordinary complex composition is exactly the zero-leakage, aligned-sector subset of the full real tensor product. The “alignment cost” is captured by a single basis-independent monotone — the **same- i leakage** $L(\rho) = \text{Tr}(P_- \rho)$ — and we develop it into a complete resource theory.

Our main quantitative results are: (i) the coherent cross-sector amplitude is bounded by $\|P_+ \rho P_-\|_1 \leq \sqrt{L(1-L)}$, saturated by pure coherent cross-sector states; (ii) a real CPTP map is leakage-nonincreasing if and only if $\Phi^\dagger(P_-) \leq P_-$, with the Kraus characterization $P_- K_k P_+ = 0$; (iii) a binary discrimination task between the identity channel and the sector-parity flip $\rho \mapsto M \rho M$ has optimal success probability $\frac{1}{2} + \sqrt{m(1-m)}$ with $m = \min\{\epsilon, 1/2\}$, achieved by coherent cross-sector probes and not by population leakage alone. The maximally real-entangled, complex-separable Caves–Fuchs–Rungta two-rebit state has $L = 0$: it sits at the center of the aligned sector, demonstrating that same- i leakage and real-entanglement detect genuinely different aspects of the real-vs-complex distinction.

We also identify a structural obstruction: for product-source bilocal networks, the natural Bob-node leakage parameter is identically $1/2$, independent of the source. This is a no-go that clarifies why network certification of complex-structure alignment requires operational, rather than product-state, source independence — a sharpening of the question raised by recent work of Hoffreumon–Woods [1, 2] and Weilenmann–Gisin–Sekatski [3].

1 Introduction

1.1 Why composition is where complex structure becomes nontrivial

Whether quantum theory is intrinsically complex-valued is one of the longest-running debates in the foundations of the subject. The recent network experiment of Renou et al. [4] sharpened it by showing that real and complex quantum theory make different predictions in bilocal scenarios under product-state source independence, and Chen et al. [5] confirmed this experimentally. Hoffreumon and Woods [1, 2] responded that a real-Hilbert-space formulation can be retained if the composition rule is changed appropriately, and Weilenmann, Gisin, and Sekatski [3] showed that partial source independence already suffices to recover the Renou conclusion. Across these positions, the live disagreement is not about whether complex amplitudes are computationally useful — they are — but about whether they are kinematically necessary once composition is treated carefully.

Our starting point is geometric. A single complex Hilbert space $H_{\mathbb{C}}$ of complex dimension d is, as a real Hilbert space, $2d$ -dimensional and equipped with an orthogonal complex structure

J satisfying $J^2 = -I$ and $J^\top = -J$. The imaginary unit is the operator J . When two such systems with local complex structures J_A, J_B are composed, the full real tensor product $K_{AB} = H_{A,\mathbb{R}} \otimes_{\mathbb{R}} H_{B,\mathbb{R}}$ has real dimension $4d_A d_B$, exactly twice that of the realified complex tensor product. The extra $2d_A d_B$ real dimensions parametrize configurations in which Alice’s “multiply by i ” and Bob’s “multiply by i ” disagree. The ordinary complex tensor product identifies them through the balancing relation

$$J_A x \otimes y = x \otimes J_B y. \quad (1)$$

Our central claim is that the failure to satisfy this relation is a **resource** in the technical sense: it admits an affine monotone, a complete classification of free operations, exact geometric bounds, and an explicit operational interpretation.

Main contributions

1. **Sector geometry.** The full real tensor product K_{AB} splits as $H_+ \oplus H_-$ under the alignment operator $M = J_A \otimes J_B$; ordinary complex composition is exactly the aligned sector H_+ together with J_{AB} -compatibility (Sections 3 and 4).
2. **Resource theory.** The same- i leakage $L(\rho) = \text{Tr}(P_- \rho)$ is an affine monotone under the exactly characterised family of channels $\Phi^\dagger(P_-) \leq P_-$, with Kraus form $P_- K_k P_+ = 0$; coherent cross-sector amplitudes are bounded by $\sqrt{L(1-L)}$ (Section 4).
3. **Operational task.** The binary discrimination between $\rho \mapsto \rho$ and $\rho \mapsto M\rho M$ at leakage budget ϵ has optimum $\frac{1}{2} + \sqrt{m(1-m)}$ with $m = \min\{\epsilon, 1/2\}$, achieved by coherent cross-sector probes (Section 7).
4. **Foundational consequence.** The Caves–Fuchs–Rungta maximally real-entangled state has zero leakage; the natural Bob-node leakage parameter in product-source bilocal networks is identically $1/2$, a structural no-go that sharpens the network certification question (Sections 3.4 and 10).

Scope

The framework quantifies, as a resource, the compositional cost of enforcing a shared complex structure across independently composed real systems. It is consistent with both directions of the Renou–Hoffreumon–Woods–Weilenmann–Gisin–Sekatski falsifiability debate: same- i leakage is a property of the full real tensor product that exists regardless of which side of that debate one takes. We do not claim that complex amplitudes are experimentally necessary, that real quantum theory is falsified, or that we obtain a non-tautological device-independent network witness. The structural distinction from Hickey–Gour basis-relative imaginarity is example-based rather than operational (Section 8).

1.2 Positioning

Our framework is distinct from four adjacent lines of work:

- The basis-relative imaginarity resource theory of Hickey–Gour [6] and Wu et al. [7, 8]: same- i leakage is basis-independent — it depends only on the choice of local complex structures J_A, J_B , not on a computational basis.
- The ubit construction of Aleksandrova–Borish–Wootters [9]: that work couples a real system to an auxiliary rebit; we decompose the bipartite real tensor product itself.
- The real-vs-complex entanglement gap of Caves–Fuchs–Rungta [10] and Chiribella et al. [11]: we recover their maximally real-entangled two-rebit state at zero leakage (Section 3.4), confirming that real entanglement and same- i leakage detect orthogonal features of the real-vs-complex distinction.

- The network falsifiability axis [4, 1, 2, 3]: that line concerns whether real and complex quantum theory are operationally distinguishable under various independence assumptions; same- i leakage quantifies a hidden compositional structure within real Hilbert space, prior to and independent of that operational question.

The framework also differs from Marvian–Spekkens asymmetry: the alignment operator M is an order-two involution whose eigenspaces partition state space, not a group generator whose orbits define a free set. This sector-grading structure is exactly what makes the standard positive-noise robustness diverge (Proposition 11). A canonical overview of resource theories is given in Chitambar–Gour [12].

1.3 Notation summary

All Hilbert spaces are finite-dimensional. The real adjoint is $\top$; in an orthonormal real basis this is transpose. Complex Hilbert spaces are $H_{\mathbb{C}}$ of complex dimension d , with realifications $H_{\mathbb{R}}$ of real dimension $2d$. The local complex structures are J_A, J_B (each satisfies $J^2 = -I$, $J^\top = -J$). The *sector operators* are

$$A := J_A \otimes I_B, \quad B := I_A \otimes J_B, \quad M := AB = J_A \otimes J_B. \quad (2)$$

Alice, Bob, and Charlie party labels appear as subscripts (ρ_{AB_1}, J_{B_1}); the symbols A, B in sans-serif are reserved for the sector operators above. We write ι for the explicit isometric embedding $H_A \otimes_{\mathbb{C}} H_B \hookrightarrow K_{AB}$ defined in Eq. (7) below. The Schrödinger real generator is $G = -JH_{\mathbb{R}}$ where $H_{\mathbb{R}}$ is the realified Hamiltonian. Lindblad jump operators are R_ℓ . The trace norm is $\|\cdot\|_1$. Density operators ρ are real symmetric positive semidefinite with $\text{Tr } \rho = 1$. The free face is $\mathcal{F}_{AB} := \{\rho : L(\rho) = 0\} = \{\rho : \rho = P_+ \rho P_+\}$.

2 Realification and local complex structures

2.1 Orthogonal complex structures

Theorem 1 (Realification preserves quantum statistics). *Let $X = A + iB$ be a $d \times d$ complex matrix with real A, B , and define*

$$R(X) := \begin{pmatrix} A & -B \\ B & A \end{pmatrix}. \quad (3)$$

Let $\rho_{\mathbb{C}}$ be a complex density operator and $E_{\mathbb{C}}$ a complex effect. Define $\rho_{\mathbb{R}} := \frac{1}{2}R(\rho_{\mathbb{C}})$ and $E_{\mathbb{R}} := R(E_{\mathbb{C}})$. Then

$$\text{Tr}_{\mathbb{R}} \rho_{\mathbb{R}} = 1, \quad \text{Tr}_{\mathbb{R}}(E_{\mathbb{R}} \rho_{\mathbb{R}}) = \text{Tr}_{\mathbb{C}}(E_{\mathbb{C}} \rho_{\mathbb{C}}). \quad (4)$$

The factor $\frac{1}{2}$ in $\rho_{\mathbb{R}} = \frac{1}{2}R(\rho_{\mathbb{C}})$ is not cosmetic: it compensates the doubled real trace and is what makes Born probabilities preserved exactly, not up to a factor of two. This is the first place where the real reformulation differs from “complex with a different bookkeeping.” The proof is in Appendix B.

The *standard* complex structure on $\mathbb{R}^{2d}$ is

$$J_d := \begin{pmatrix} 0 & -I_d \\ I_d & 0 \end{pmatrix}, \quad (5)$$

satisfying $J_d^2 = -I$ and $J_d^\top = -J_d$. The induced complex inner product is $\langle \psi, \phi \rangle_J = g(\psi, \phi) + i g(J\psi, \phi)$, where g is the real inner product. A real-linear operator X on $(H_{\mathbb{R}}, J)$ is complex-linear iff $XJ = JX$.

3 Same- i composition

3.1 The quotient theorem

Suppose Alice prepares a state in $H_{A, \mathbb{R}}$ and Bob prepares a state in $H_{B, \mathbb{R}}$, each working with their own choice of local complex structure J_A, J_B . What does it mean for the joint state to “be a

complex state”? The complex tensor product imposes the balancing relation $J_A x \otimes y = x \otimes J_B y$ — that is, multiplying Alice’s slot by i produces the same vector as multiplying Bob’s slot by i . Without this relation, the joint real tensor product accommodates configurations in which the two parties’ notions of i are inconsistent. We show that the balancing relation is, equivalently, the projection onto a specific eigenspace of the operator $M = J_A \otimes J_B$, and that this eigenspace is isomorphic to the complex tensor product. The complementary eigenspace is the **leakage sector**: it has the same real dimension as the aligned sector and represents anti-aligned configurations.

Theorem 2 (Same- i quotient). *Let $H_{\mathbb{C}}^A, H_{\mathbb{C}}^B$ be complex Hilbert spaces of complex dimensions d_A, d_B , with realifications $H_{A,\mathbb{R}}, H_{B,\mathbb{R}}$ carrying complex structures J_A, J_B . There is a real-linear isomorphism*

$$(H_{\mathbb{C}}^A \otimes_{\mathbb{C}} H_{\mathbb{C}}^B)_{\mathbb{R}} \cong \frac{H_{A,\mathbb{R}} \otimes_{\mathbb{R}} H_{B,\mathbb{R}}}{\text{span}_{\mathbb{R}}\{J_A x \otimes y - x \otimes J_B y : x \in H_{A,\mathbb{R}}, y \in H_{B,\mathbb{R}}\}}. \quad (6)$$

Proof. The complex tensor product is the universal target of complex-bilinear maps. A real-bilinear map $\varphi : H_{A,\mathbb{R}} \times H_{B,\mathbb{R}} \rightarrow V$ into a complex vector space V factors through the complex tensor product iff $\varphi(J_A x, y) = \varphi(x, J_B y)$ for all x, y . Equivalently, φ vanishes on the balancing relations $J_A x \otimes y - x \otimes J_B y$. The quotient therefore satisfies the universal property of $(H_{\mathbb{C}}^A \otimes_{\mathbb{C}} H_{\mathbb{C}}^B)_{\mathbb{R}}$. $\square$

3.2 Sector decomposition and the embedding ι

Define $A = J_A \otimes I_B$, $B = I_A \otimes J_B$, and $M = AB = J_A \otimes J_B$. Then M is a real symmetric involution: $M^{\top} = M$ and $M^2 = I$. The real tensor product $K_{AB} = H_{A,\mathbb{R}} \otimes_{\mathbb{R}} H_{B,\mathbb{R}}$ has $\dim_{\mathbb{R}} K_{AB} = 4d_A d_B$. Define $H_+ = \ker(M + I)$, $H_- = \ker(M - I)$, and $P_+ = \frac{1}{2}(I - M)$, $P_- = \frac{1}{2}(I + M)$.

We will use the explicit isometric embedding

$$\iota : H_A \otimes_{\mathbb{C}} H_B \hookrightarrow K_{AB}, \quad \iota(x \otimes_{\mathbb{C}} y) = \frac{1}{\sqrt{2}}(x \otimes y - J_A x \otimes J_B y). \quad (7)$$

A direct calculation shows ι is real-linear, isometric, has image in H_+ , and intertwines the complex structures: $\iota(i \cdot x \otimes_{\mathbb{C}} y) = J_{AB} \iota(x \otimes_{\mathbb{C}} y)$.

Theorem 3 (Sector representation of the quotient). *The aligned sector H_+ represents the same- i quotient; equivalently, $\psi \in H_+$ iff $A\psi = B\psi$. The sector has real dimension $\dim_{\mathbb{R}} H_+ = 2d_A d_B$ and carries the induced complex structure*

$$J_{AB} := A|_{H_+} = B|_{H_+}, \quad J_{AB}^2 = -I_{H_+}. \quad (8)$$

Proof. Proof sketch. Applying A to $A\psi = B\psi$ gives $A^2\psi = AB\psi = M\psi$, and $A^2 = -I$ forces $M\psi = -\psi$. Conversely, $M\psi = -\psi$ gives $B\psi = -A^{-1}\psi = A\psi$. Dimension count: $\text{Tr } M = \text{Tr}(J_A) \text{Tr}(J_B) = 0$ (each J antisymmetric) forces $\dim H_+ = \dim H_- = 2d_A d_B$. $\square$

Figure 1 illustrates the sector decomposition.

3.3 Recovery of ordinary complex quantum theory

A real density operator ρ on K_{AB} corresponds to a state of ordinary complex quantum theory on $H_{\mathbb{C}}^A \otimes_{\mathbb{C}} H_{\mathbb{C}}^B$ iff

$$\rho = P_+ \rho P_+ \quad \text{and} \quad [\rho, J_{AB}] = 0. \quad (9)$$

The first condition is zero leakage, $L(\rho) = 0$; the second is J_{AB} -compatibility on the aligned sector. These are independent constraints. The second condition is a unitarity constraint at fixed sector; once a state sits inside H_+ it is a unitarily evolved complex state. The compositional question — whether the state is in H_+ at all — is logically prior, and is the question we address here.

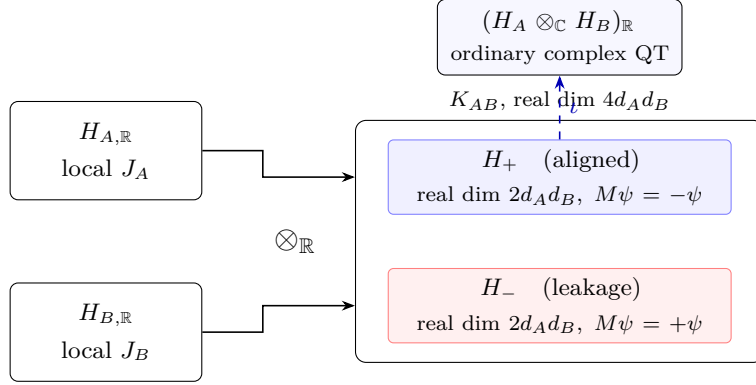


Figure 1: Geometric structure of same- i composition. Two real Hilbert spaces $H_{A,\mathbb{R}}, H_{B,\mathbb{R}}$ equipped with local complex structures J_A, J_B combine via the real tensor product into K_{AB} of real dimension $4d_A d_B$. The alignment operator $M = J_A \otimes J_B$ decomposes K_{AB} into orthogonal sectors H_+, H_- of equal real dimension $2d_A d_B$. The aligned sector H_+ represents ordinary complex composition; the isometric embedding ι (Eq. (7)) identifies it with $(H_A \otimes_{\mathbb{C}} H_B)_{\mathbb{R}}$. States with support in H_- — the leakage sector — describe configurations in which the two subsystems' local notions of multiplication by i disagree.

3.4 Worked example: the Caves–Fuchs–Rungta two-rebit family

To make the sector decomposition concrete, we apply it to the two-rebit family introduced by Caves, Fuchs, and Rungta [10]:

$$\rho_{\text{CFR}}(\alpha) = \frac{1}{4}(\sigma_0 \otimes \sigma_0 + \alpha \sigma_y \otimes \sigma_y), \quad -1 \leq \alpha \leq 1. \quad (10)$$

This state is real-entangled with real concurrence $|\alpha|$ but complex-separable. Each σ_y is purely imaginary as a complex matrix, but $\sigma_y \otimes \sigma_y$ has real entries, so $\rho_{\text{CFR}}(\alpha)$ is a bona fide real density operator. Taking $J_A = J_B = i\sigma_y$ as the standard 2D real complex structure, $M = J_A \otimes J_B = -\sigma_y \otimes \sigma_y$, so $\rho_{\text{CFR}}(\alpha) = \frac{1}{4}(I - \alpha M)$.

Proposition 4 (CFR leakage). $L(\rho_{\text{CFR}}(\alpha)) = (1 - \alpha)/2$.

Proof. $L = \text{Tr}(P_- \rho_{\text{CFR}}(\alpha)) = \frac{1}{8} \text{Tr}((I + M)(I - \alpha M)) = \frac{1}{8} \text{Tr}(I - \alpha M + M - \alpha M^2) = \frac{1}{8}(4 - 4\alpha) = (1 - \alpha)/2$, using $\text{Tr} M = 0$ (each $i\sigma_y$ is traceless and antisymmetric), $M^2 = I$, and $\text{Tr} I_4 = 4$. $\square$

At $\alpha = 1$ — the maximally real-entangled CFR state — the leakage vanishes: $\rho = \frac{1}{2}P_+$, sitting at the center of the aligned sector. This is the first nontrivial physical fact about same- i leakage: *the standard real-entangled, complex-separable state is fully aligned*. Real entanglement and same- i leakage detect different things. At $\alpha = -1$, leakage is 1: fully anti-aligned. Figure 2 shows the linear interpolation.

4 Same- i leakage as a resource

4.1 Definition and basic properties

The **same- i leakage** of a real density operator ρ on K_{AB} is

$$L(\rho) := \text{Tr}(P_- \rho) = \frac{1}{2}(1 + \text{Tr}(M\rho)). \quad (11)$$

The **free face** is $\mathcal{F}_{AB} = \{\rho : L(\rho) = 0\}$. Manifestly L is affine in ρ and bounded: $0 \leq L \leq 1$, with $L = 0$ if and only if $\rho = P_+ \rho P_+$ — that is, ρ is supported entirely in the aligned sector. (The non-trivial direction uses $\text{Tr}(P_- \rho P_-) = 0$ together with $P_- \rho P_- \succeq 0$, forcing $P_- \rho P_- = 0$; then $\rho \succeq 0$ on the bipartite block decomposition forces the off-diagonal blocks $\rho_{+-} = 0$ by Schur complement.)

The leakage $L(\rho)$ measures the probability weight outside the aligned sector — the chance that a sector-parity measurement returns the anti-aligned outcome. It is insensitive to coherence

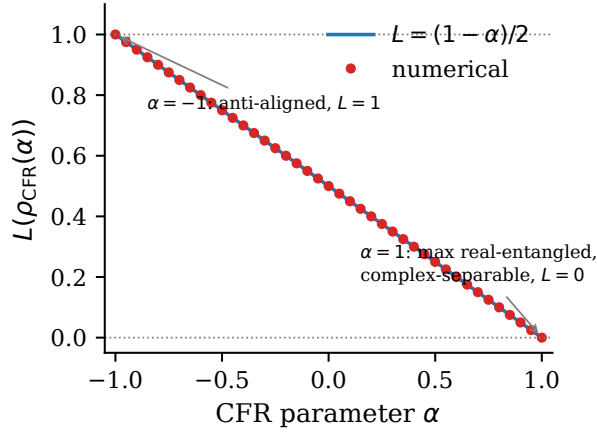


Figure 2: Leakage of the Caves–Fuchs–Rungta family $\rho_{\text{CFR}}(\alpha)$ as a function of α . The closed-form prediction $L = (1 - \alpha)/2$ from Proposition 4 (solid curve) matches numerical evaluation of $L = \text{Tr}(P_- \rho_{\text{CFR}}(\alpha))$ (points). At $\alpha = 1$ (max real-entangled, complex-separable) leakage vanishes; at $\alpha = -1$ (anti-aligned) leakage equals 1.

between sectors. That coherence is captured separately by the **coherent leakage**

$$C(\rho) := \|P_+ \rho P_-\|_1, \quad (12)$$

which is the operationally relevant quantity for the discrimination task in Section 7, and which is constrained by L through the bound below.

Theorem 5 (Coherent leakage bound). *For every real density operator ρ ,*

$$\|P_+ \rho P_-\|_1 \leq \sqrt{L(\rho)(1 - L(\rho))}, \quad (13)$$

with equality on pure coherent cross-sector states.

Proof sketch. Douglas factorization for PSD block matrices [13] writes $\rho_{+-} = \rho_{++}^{1/2} K \rho_{--}^{1/2}$ with $\|K\|_\infty \leq 1$; Hölder then gives $\|\rho_{+-}\|_1 \leq \sqrt{\text{Tr} \rho_{++}} \sqrt{\text{Tr} \rho_{--}} = \sqrt{(1 - L)\bar{L}}$. Saturation by $|\psi\rangle = \sqrt{1 - \lambda}|\psi_+\rangle + \sqrt{\lambda}|\psi_-\rangle$ is direct. See Appendix B for details. $\square$

4.2 Free operations

Theorem 6 (Leakage monotonicity criterion). *A real CPTP map Φ on K_{AB} satisfies $L(\Phi(\rho)) \leq L(\rho)$ for every density operator ρ iff $\Phi^\dagger(P_-) \leq P_-$.*

(Forward direction by duality: $L(\Phi(\rho)) - L(\rho) = \text{Tr}((\Phi^\dagger(P_-) - P_-)\rho)$. Full proof in Appendix B.)

Theorem 7 (Kraus criterion for leakage-nonincreasing channels). *For a real CP map $\Phi(\rho) = \sum_k K_k \rho K_k^\top$ satisfying $\sum_k K_k^\top K_k = I$, the condition $\Phi^\dagger(P_-) \leq P_-$ holds iff*

$$P_- K_k P_+ = 0 \quad \text{for every } k. \quad (14)$$

(Proof in Appendix B.)

Definition 8 (Strongly sector-preserving channels). A real CPTP channel Φ with Kraus operators $\{K_k\}$ is *strongly sector-preserving* if $[K_k, M] = 0$ for every k . Strongly sector-preserving channels conserve leakage exactly: $L(\Phi(\rho)) = L(\rho)$.

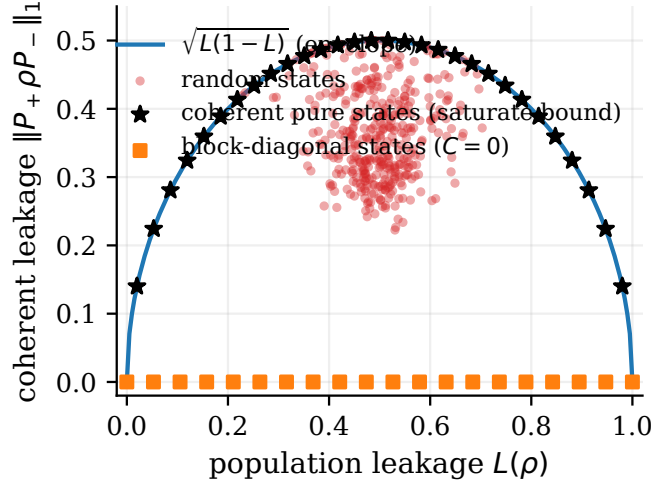


Figure 3: Coherent leakage $\|P_+ \rho P_-\|_1$ against population leakage $L(\rho)$. Solid curve: saturation envelope $\sqrt{L(1-L)}$ from Theorem 5. Red dots: 400 random real density operators of varying rank on K_{AB} with $d_A = d_B = 2$. Black stars: pure coherent cross-sector states, which saturate the bound. Orange squares: block-diagonal states, which lie on the $C = 0$ axis regardless of L .

5 Geometry of the resource

5.1 Trace distance to the free face on coherent pure states

Theorem 9 (Trace distance to the free face: coherent pure states). *Let $|\psi\rangle = \sqrt{1-\lambda}|\psi_+\rangle + \sqrt{\lambda}|\psi_-\rangle$ with $|\psi_\pm\rangle \in H_\pm$ unit and $0 \leq \lambda \leq 1$. Then*

$$D_1(|\psi\rangle\langle\psi|, \mathcal{F}_{AB}) := \inf_{\sigma \in \mathcal{F}_{AB}} \frac{1}{2} \|\psi\rangle\langle\psi| - \sigma\|_1 = \sqrt{\lambda}. \quad (15)$$

Proof sketch. Upper bound: $\sigma = |\psi_+\rangle\langle\psi_+| \in \mathcal{F}_{AB}$ gives $\frac{1}{2} \|\psi\rangle\langle\psi| - \sigma\|_1 = \sqrt{1 - |\langle\psi|\psi_+\rangle|^2} = \sqrt{\lambda}$. The matching lower bound is achieved via an explicit dual witness Y on $\text{span}\{\psi_+, \psi_-\}$; see Appendix B. $\square$

5.2 Alignment weight

The convex-roof alignment weight is

$$W(\rho) := \inf\{\lambda \in [0, 1] : \rho = (1-\lambda)\sigma + \lambda\tau, \sigma \in \mathcal{F}_{AB}, \tau \in \mathcal{D}(K_{AB})\}. \quad (16)$$

Proposition 10 (Alignment weight ordering). *$W(\rho) \geq L(\rho)$, with equality on states block-diagonal in $H_+ \oplus H_-$. For coherent pure cross-sector states with $0 < \lambda < 1$, $W = 1$ while $L = \lambda$.*

Proof. If $\rho = (1-\lambda)\sigma + \lambda\tau$ with σ free, then $L(\rho) = \lambda L(\tau) \leq \lambda$; taking the infimum gives $L \leq W$. For block-diagonal ρ , the decomposition $\rho = (\rho_{++} \oplus 0) + (0 \oplus \rho_{--})$ realizes $W = L$. For a coherent pure state, extremality forces the decomposition to be trivial ($\sigma = |\psi\rangle\langle\psi|$ is not free), so $W = 1$. $\square$

5.3 Divergent positive-noise generalized robustness

Proposition 11 (Divergent positive-noise generalized robustness). *The standard generalized robustness*

$$R(\rho) := \inf\{s \geq 0 : (\rho + s\tau)/(1+s) \in \mathcal{F}_{AB}, \tau \in \mathcal{D}(K_{AB})\} \quad (17)$$

is infinite whenever $L(\rho) > 0$.

Proof. If $(\rho + s\tau)/(1+s) \in \mathcal{F}_{AB}$, then $P_-(\rho + s\tau)P_- = 0$. Both $P_-\rho P_-$ and $P_-\tau P_-$ are PSD, and their sum vanishes only if each vanishes. Hence $L(\rho) = 0$. $\square$

The divergence is structural: it reflects that $\mathcal{F}_{AB}$ is a proper face of the state space (not an interior region of a convex cone of “noisy free states”), and is therefore generic for sector-grading resource theories. It also means that the standard Takagi–Regula [14] robustness-quantifies-advantage framework does not directly apply; the operational quantifier in our setting is trace distance, not robustness.

5.4 Object summary

Object	Formula	Physical meaning
J_A, J_B	$J^2 = -I, J^\top = -J$	Local complex structures
M	$J_A \otimes J_B$	Alignment operator
$H_\pm$	$\ker(M \pm I)$	Aligned / anti-aligned sectors
$P_\pm$	$\frac{1}{2}(I \mp M)$	Sector projectors
$L(\rho)$	$\text{Tr}(P_- \rho)$	Population leakage
$C(\rho)$	$\ P_+ \rho P_- \ _1$	Coherent leakage
$W(\rho)$	convex-roof alignment weight	Mixture resource
$D_1(\rho, \mathcal{F}_{AB})$	$\inf_{\sigma \in \mathcal{F}_{AB}} \frac{1}{2} \ \rho - \sigma\ _1$	Trace distance to free face
$R(\rho)$	generalized robustness	$= \infty$ for $L > 0$

6 Dynamics

Up to this point the framework has been static. We now ask: what dynamics preserve, generate, or destroy same- i leakage? A Hamiltonian generator G that commutes with M produces leakage-conserving evolution; perturbations breaking that commutation generate leakage at quadratic order in the perturbation strength; and dissipative Lindblad evolutions admit an explicit rate formula whose vanishing is characterised by a simple commutator condition on the jump operators. These results convert the abstract sector structure into a usable physical predictor: given a real-Hilbert-space model with explicit dynamics, one can check whether leakage is conserved by inspection.

6.1 Hamiltonian leakage

Let $H_{\mathbb{R}}$ be a real symmetric Hamiltonian on K_{AB} . The Schrödinger generator is $G = -JH_{\mathbb{R}}$, which is skew-symmetric ($G^\top = -G$), so the evolution $O(t) = e^{Gt}$ is orthogonal.

Theorem 12 (Hamiltonian leakage conservation). *If $[G, M] = 0$, then $L(\rho(t)) = L(\rho(0))$ for all t .*

Proof. $[G, M] = 0$ implies $[e^{Gt}, P_-] = 0$, so $\text{Tr}(P_- e^{Gt} \rho e^{-Gt}) = \text{Tr}(P_- \rho)$. $\square$

6.2 Perturbative leakage generation

Theorem 13 (Perturbative leakage). *Let $G_\epsilon = G_0 + \epsilon B$ with $G_0^\top = -G_0$, $B^\top = -B$, and $[G_0, M] = 0$. For $\psi_0 \in H_+$ unit and finite $t > 0$,*

$$L(|e^{G_\epsilon t} \psi_0\rangle\langle e^{G_\epsilon t} \psi_0|) = \epsilon^2 \left\| \int_0^t P_- e^{G_0(t-s)} B e^{G_0 s} \psi_0 ds \right\|^2 + O(\epsilon^3). \quad (18)$$

(Duhamel-expansion proof in Appendix B.)

6.3 Lindblad leakage rate

A real Lindblad generator and its dual are

$$\mathcal{L}(\rho) = G\rho + \rho G^\top + \sum_\ell (R_\ell \rho R_\ell^\top - \frac{1}{2} \{R_\ell^\top R_\ell, \rho\}), \quad \mathcal{L}^\dagger(X) = [X, G] + \sum_\ell (R_\ell^\top X R_\ell - \frac{1}{2} \{R_\ell^\top R_\ell, X\}). \quad (19)$$

Theorem 14 (Lindblad leakage rate). $\frac{d}{dt}L(\rho_t) = \text{Tr}(\mathcal{L}^\dagger(P_-)\rho_t)$. If $\mathcal{L}^\dagger(P_-) = 0$, leakage is conserved; if $\mathcal{L}^\dagger(P_-) \leq 0$, leakage is nonincreasing.

Corollary 15 (Sufficient conservation condition). If $[G, M] = 0$ and $[R_\ell, M] = 0$ for every ℓ , then $\mathcal{L}^\dagger(P_-) = 0$ and $L(\rho_t) = L(\rho_0)$ for all t .

Proof. $[G, M] = 0$ implies $[P_-, G] = 0$, so the Hamiltonian-commutator term in $\mathcal{L}^\dagger(P_-)$ vanishes. For each dissipator, $[R_\ell, M] = 0$ gives $[R_\ell, P_-] = 0$ and $[R_\ell^\top, P_-] = 0$, hence $R_\ell^\top R_\ell$ commutes with P_- . Then $R_\ell^\top P_- R_\ell = R_\ell^\top R_\ell P_- = \frac{1}{2}\{R_\ell^\top R_\ell, P_-\}$, so each dissipator term vanishes. $\square$

7 Sector-parity discrimination

So far the framework has been structural and dynamical; we now extract an operational task. Suppose a verifier hands Alice a black-box channel that is either the identity $\mathcal{N}_0(\rho) = \rho$ or the sector-parity flip $\mathcal{N}_1(\rho) = M\rho M$, with equal prior. Alice prepares a probe state ρ whose leakage is bounded by ϵ , sends it through the box, and measures the output. What is her optimal probability of correctly identifying the channel?

This task is not a network witness — there is no spatial separation involved, no source-independence requirement, and the channel $M\rho M$ is constructed by hand rather than emerging from a physical scenario. Rather, it is the simplest internal operational probe of the resource theory: a “minimum-effort” task whose closed-form optimum demonstrates how leakage and coherent cross-sector amplitude translate into distinguishability. The optimal strategy involves coherent cross-sector probes; pure population leakage is operationally inert.

Theorem 16 (Sector-parity discrimination). *The optimal binary discrimination success probability with equal priors and leakage budget $L(\rho) \leq \epsilon$ is*

$$p_{\text{succ}}^{(\epsilon)} = \frac{1}{2} + \sqrt{m(1-m)}, \quad m = \min\{\epsilon, 1/2\}. \quad (20)$$

Equivalently, $p_{\text{succ}}^{(\epsilon)} = \frac{1}{2} + \sqrt{\epsilon(1-\epsilon)}$ for $0 \leq \epsilon \leq 1/2$, and $p_{\text{succ}}^{(\epsilon)} = 1$ for $\epsilon \geq 1/2$, saturated by coherent pure cross-sector states.

Proof. The matrix $X := P_+\rho P_- + P_-\rho P_+$ has off-diagonal block structure with $B = P_+\rho P_-$; its eigenvalues come in pairs $\pm\sigma_i(B)$, so $\|X\|_1 = 2\|B\|_1$. Hence $\|\rho - M\rho M\|_1 = 4\|P_+\rho P_-\|_1$, and the Holevo–Helstrom theorem [15] gives $p_{\text{succ}}(\rho) = \frac{1}{2} + \|P_+\rho P_-\|_1$. Maximising over $\{\rho : L(\rho) \leq \epsilon\}$ using Theorem 5: the function $L \mapsto \sqrt{L(1-L)}$ is increasing on $[0, 1/2]$ and peaks at $L = 1/2$. For $\epsilon < 1/2$ the constraint is tight and the optimum is at $L = \epsilon$; for $\epsilon \geq 1/2$ the unconstrained optimum $L = 1/2$ satisfies the budget. The bound is saturated by the pure coherent cross-sector state $|\psi\rangle = \sqrt{1-m}|\psi_+\rangle + \sqrt{m}|\psi_-\rangle$ with $m = \min\{\epsilon, 1/2\}$. $\square$

Block-diagonal probes give no advantage. For ρ block-diagonal on $H_+ \oplus H_-$, $P_+\rho P_- = 0$ and $p_{\text{succ}}(\rho) = 1/2$. Population leakage alone provides no advantage; only coherent cross-sector amplitude does.

Limitation: imaginarity is not separated by this task. The optimal probe can be chosen with entrywise-real entries in the basis where J_A, J_B are the standard symplectic matrices: take $\psi_+, \psi_- \in H_\pm$ unit and entrywise real (always possible since $H_\pm$ are real subspaces); then $|\psi\rangle\langle\psi|$ is entrywise real and has Hickey–Gour imaginarity zero in that basis, yet it achieves the maximal success probability.

8 Structural distinction from basis imaginarity

The Hickey–Gour resource theory of imaginarity [6, 7] defines a state as “free” if its density matrix has only real entries in a fixed reference basis. This is a *basis-relative* notion. Same- i leakage, by contrast, depends only on the local complex structures J_A, J_B and is basis-independent. The

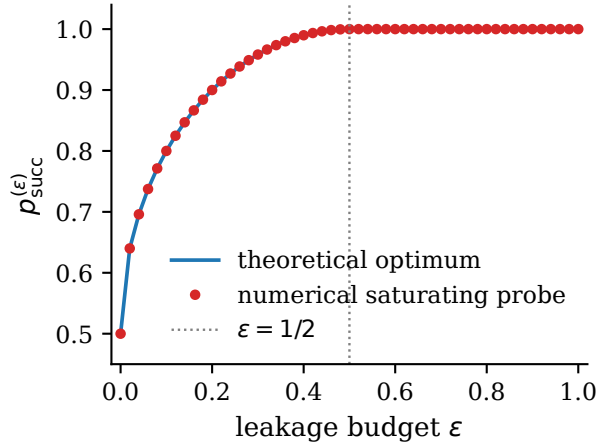


Figure 4: Sector-parity optimal success probability as a function of leakage budget ϵ . Solid curve: theoretical optimum $\frac{1}{2} + \sqrt{m(1-m)}$, $m = \min\{\epsilon, 1/2\}$. Points: numerical saturating probe. The vertical line at $\epsilon = 1/2$ marks where the optimum reaches 1.

two are therefore distinct resources, but the precise nature of the distinction is example-based, not operational. We illustrate this with two extremal cases. Åberg [16] introduced the closely related superposition quantification framework that motivates both notions.

Example 17 (Entrywise real, $L = 1/2$). Let $\psi_{\pm} \in H_{\pm}$ be entrywise-real unit vectors (possible because $H_{\pm}$ are real subspaces). Take $\psi = (\psi_{+} + \psi_{-})/\sqrt{2}$. Then $\rho = |\psi\rangle\langle\psi|$ has entirely real entries in the standard basis — so its Hickey–Gour imaginarity is zero in that basis — but $L(\rho) = 1/2$.

Example 18 ($L = 0$, nonzero entrywise imaginarity). Let $|\psi\rangle_{\mathbb{C}} = (|00\rangle + i|11\rangle)/\sqrt{2}$ on $\mathbb{C}^2 \otimes_{\mathbb{C}} \mathbb{C}^2$. Its density operator has nonzero imaginary entries in the computational basis. Apply ι from Eq. (7): $\iota(|\psi\rangle_{\mathbb{C}})$ is a unit vector in $H_{+} \subset K_{AB}$, and $\rho_R = |\iota(|\psi\rangle_{\mathbb{C}})\rangle\langle\iota(|\psi\rangle_{\mathbb{C}})|$ satisfies $L(\rho_R) = 0$ by construction.

Examples 17 and 18 demonstrate that no map $L \mapsto \mathcal{I}_B$ and no map $\mathcal{I}_B \mapsto L$ exists that is consistent across the standard real basis: the two measures are structurally distinct. Whether there exists an operational task — a channel pair and probe class — for which L -advantage cannot be achieved by any zero-imaginarity state in any basis is open. We conjecture that no such task exists for two-rebit channels, and that the simplest separating task requires composition of $d_A = d_B = 2$ systems with non-trivial entanglement structure; this is a natural follow-up direction.

9 The Caves–Fuchs–Rungta family revisited

We computed in Section 3.4 that $L(\rho_{\text{CFR}}(\alpha)) = (1 - \alpha)/2$. We extract two further geometric statements that this family makes vivid.

Position in the coherent-leakage diagram. Since $\rho_{\text{CFR}}(\alpha) = \frac{1}{4}(I - \alpha M)$ is block-diagonal in $H_{+} \oplus H_{-}$ (both the identity and M are block-diagonal in this decomposition), its coherent leakage $C(\rho_{\text{CFR}}(\alpha)) = 0$ for all α . The CFR family therefore lies on the $C = 0$ axis of Figure 3, regardless of α .

Perturbative dynamics. Applying the perturbative leakage formula (Theorem 13) to an initial CFR state $\rho_0 = \rho_{\text{CFR}}(1) = \frac{1}{2}P_{+}$ under a perturbation that breaks $[G_0, M] = 0$: since ρ_0 is supported in H_{+} , any first-order Duhamel contribution is killed by P_{-} , and leakage grows as $O(\epsilon^2)$, consistent with the block-diagonal structure.

Connection to the ubit construction. Aleksandrova–Borish–Wootters [9] couple a real system to an auxiliary rebit (“ubit”) to recover complex predictions. The CFR two-rebit state at $\alpha = 1$ corresponds to maximal alignment between Alice’s and Bob’s i -structures, which is precisely the condition the ubit construction enforces by hand. Same- i leakage quantifies how far a general state deviates from that enforced alignment.

10 A structural obstruction to product-source bilocal witnesses

Same- i leakage is a property of an actual real-Hilbert-space state. To turn it into a network witness — a statement that some experimentally observable correlation cannot be reproduced by any real-source bilocal model — one would naively define a Bob-node leakage parameter L_B on the reduced state $\rho_{B_1 B_2}$ and look for sources with $L_B > 1/2$. We now show that this naive programme fails for a structural reason: for every *product* bilocal source $\rho_{B_1 B_2} = \rho_{B_1} \otimes \rho_{B_2}$, the Bob-node leakage is identically $1/2$. This is not a contingent failure; it is a no-go theorem with a one-line proof.

Consider the bilocal scenario $A \leftarrow S_1 \rightarrow B \leftarrow S_2 \rightarrow C$, with independent product sources $\rho_{\text{net}} = \rho_{AB_1} \otimes_{\mathbb{R}} \rho_{B_2 C}$. The natural Bob-node alignment operator is $M_B = J_{B_1} \otimes J_{B_2}$, and $L_B(\rho_{B_1 B_2}) := \text{Tr}(P_-^B \rho_{B_1 B_2})$ with $P_-^B = \frac{1}{2}(I + M_B)$.

Proposition 19 (Bob-node obstruction). *For every product bilocal source $\rho_{B_1 B_2} = \rho_{B_1} \otimes_{\mathbb{R}} \rho_{B_2}$ with ρ_{B_i} real symmetric PSD,*

$$L_B(\rho_{B_1 B_2}) = \frac{1}{2} \quad \text{exactly.} \quad (21)$$

Proof. $\text{Tr}(M_B \rho_{B_1} \otimes \rho_{B_2}) = \text{Tr}(J_{B_1} \rho_{B_1}) \cdot \text{Tr}(J_{B_2} \rho_{B_2})$. Real symmetric ρ and real antisymmetric J satisfy $\text{Tr}(J\rho) = \text{Tr}((J\rho)^\top) = \text{Tr}(\rho^\top J^\top) = -\text{Tr}(J\rho)$, hence $\text{Tr}(J\rho) = 0$. Each factor vanishes, so $L_B = \frac{1}{2}(1 + 0) = \frac{1}{2}$. $\square$

The obstruction has two consequences, one negative and one positive. **Negative:** no quantity built linearly from the marginal $\rho_{B_1 B_2}$ in a product bilocal model can detect alignment, because the linear functional $\text{Tr}(M_B \cdot)$ vanishes identically. Network certification of same- i leakage cannot proceed via the most direct route. **Positive:** the obstruction identifies precisely which feature of the bilocal scenario must be relaxed for a non-trivial witness to exist. Three natural alternatives are: (i) replacing product-state with operational source independence in the sense of Hoffreumon–Woods [1, 2]; (ii) constructing the witness from a non-product correlation functional involving Alice’s and Charlie’s measurement settings that survives the Bob-node marginalisation; (iii) restricting to J -covariant sources, in which case the algebraic identity producing $1/2$ no longer holds. The obstruction makes precise what work remains.

11 Discussion

11.1 What does same- i leakage teach us about real vs. complex composition?

Ordinary complex composition is exactly the zero-leakage, aligned sector of a strictly larger real tensor product. Real Hilbert space accommodates more than complex Hilbert space in bipartite systems, and the “more” is a well-defined resource with affine monotone, exactly characterised free operations, and operational meaning via the sector-parity discrimination task. The divergence of the standard positive-noise robustness (Proposition 11) reflects that the free face is a proper face, not a symmetry orbit, of the state space.

11.2 Why is same- i leakage not just imaginarity?

Imaginarity is basis-relative; same- i leakage is basis-independent. Examples 17 and 18 establish structural inequivalence. Operational separation remains open and is conjectured to require $d_A = d_B = 2$ systems with non-trivial entanglement structure. The sector-parity discrimination task can be saturated by entrywise-real probes (Section 7), so it does not operationally separate the two resources.

11.3 Why do product-source bilocal network witnesses fail?

Proposition 19 is the structural answer: the linear functional $\text{Tr}(M_B \cdot)$ vanishes identically on product real density operators. Three alternative routes exist — operational independence, non-product correlation functionals, J -covariant sources — each open.

11.4 What is still missing for a stronger foundations claim?

A non-tautological device-independent network certificate that uses operational rather than product-state source independence, plus its connection to the Hoffreumon–Woods [1, 2] and Weilenmann–Gisin–Sekatski [3] lines of work. The framework extends to N parties via the multiparty alignment projector $P_{\text{phys}} = \prod_{i=2}^N \frac{1}{2}(I - A_i A_i)$; the multiparty monotone, its dynamics, and its operational tasks are open. Asymptotic conversion rates between coherent and population leakage, and distillation under free operations, are also open.

12 Reproducibility

A companion code repository implements the constructions of this paper as a Python package and includes a pytest suite of 59 numerical sanity checks. These checks confirm the algebraic identities of Section 3 (sector projectors, the embedding ι , the alignment trace identity), the inequality of Theorem 5 on random PSD states (4-, 8-, and 16-dimensional, 10^4 trials each), the perturbative scaling of Theorem 13, the conservation condition of Corollary 15, the closed-form CFR leakage of Proposition 4, the Bob-node obstruction of Proposition 19 on product sources, and the discrimination optimum of Theorem 16. Tolerances are 10^{-10} for algebraic identities and 10^{-3} for Euler-integrated Lindblad dynamics. All figures are generated by `figures/make_figures.py`.

The repository is distributed under the MIT License at https://github.com/JulianAttemptsCoding/complex_quantum_compute_theory. We do not claim the code verifies theorems; the code checks that the theorems’ predictions match numerical computation on specific examples.

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A Theorem inventory

Label	Statement	Location
Theorem 1	Realification preserves statistics, $\rho_{\mathbb{R}} = \frac{1}{2}R(\rho_{\mathbb{C}})$.	§2, proof App. B
Theorem 2	Same- i quotient: $(H_{\mathbb{C}}^A \otimes_{\mathbb{C}} H_{\mathbb{C}}^B)_{\mathbb{R}} \cong K_{AB} / \text{span}\{J_A x \otimes y - x \otimes J_B y\}$.	§3
Theorem 3	Aligned sector $H_+ = \ker(M + I)$ carries induced complex structure J_{AB} .	§3.2
Proposition 4	CFR leakage $L(\rho_{\text{CFR}}(\alpha)) = (1 - \alpha)/2$.	§3.4
Def. 8	Same- i leakage $L(\rho) := \text{Tr}(P_- \rho)$; free face $\mathcal{F}_{AB}$.	§4
Theorem 5	Coherent leakage bound $\ P_+ \rho P_- \ _1 \leq \sqrt{L(1 - L)}$.	§4, proof App. B
Theorem 6	Monotonicity iff $\Phi^\dagger(P_-) \leq P_-$.	§4, proof App. B
Theorem 7	Kraus criterion for leakage-nonincreasing: $P_- K_k P_+ = 0$.	§4, proof App. B
Theorem 9	$D_1(\psi\rangle\langle\psi , \mathcal{F}_{AB}) = \sqrt{\lambda}$ on coherent pure states.	§5, proof App. B
Proposition 10	$W \geq L$; $W = L$ block-diagonally; $W = 1$ on coherent pure.	§5
Proposition 11	Divergent positive-noise generalized robustness: $R = \infty$ when $L > 0$.	§5
Theorem 12	Hamiltonian leakage conservation when $[G, M] = 0$.	§6
Theorem 13	Perturbative leakage $O(\epsilon^2)$.	§6, proof App. B
Theorem 14	Lindblad rate $\dot{L} = \text{Tr}(\mathcal{L}^\dagger(P_-)\rho_t)$.	§6
Corollary 15	$[G, M] = 0 = [R_\ell, M]$ for all ℓ implies exact Lindblad conservation.	§6
Theorem 16	Sector-parity discrimination optimum $\frac{1}{2} + \sqrt{m(1 - m)}$.	§7
Proposition 19	Bob-node obstruction $L_B = 1/2$ for product bilocal sources.	§10
Theorem 20	Moment-matrix soundness (SDP oracle).	App. C

B Proofs of main results

Proof of Theorem 1

The map R is a unital ring homomorphism: $R(XY) = R(X)R(Y)$ by direct block computation. For $X = A + iB$, $\text{Tr}_{\mathbb{R}} R(X) = 2 \text{Tr} A = 2\Re \text{Tr}_{\mathbb{C}} X$. Since $\rho_{\mathbb{C}}$ is Hermitian, $\text{Tr} \rho_{\mathbb{C}} \in \mathbb{R}$ and equals 1, so $\text{Tr}_{\mathbb{R}} R(\rho_{\mathbb{C}}) = 2$, giving $\text{Tr}_{\mathbb{R}} \rho_{\mathbb{R}} = 1$. Therefore

$$\text{Tr}_{\mathbb{R}}(E_{\mathbb{R}}\rho_{\mathbb{R}}) = \frac{1}{2} \text{Tr}_{\mathbb{R}} R(E_{\mathbb{C}}\rho_{\mathbb{C}}) = \Re \text{Tr}_{\mathbb{C}}(E_{\mathbb{C}}\rho_{\mathbb{C}}) = \text{Tr}_{\mathbb{C}}(E_{\mathbb{C}}\rho_{\mathbb{C}}),$$

where the last equality uses that the trace of a product of Hermitian operators is real. $\square$

Proof of Theorem 5 (Coherent leakage bound)

By the Douglas factorization for PSD block matrices [13], there exists a contraction K with $\|K\|_{\infty} \leq 1$ such that $\rho_{+-} = \rho_{++}^{1/2} K \rho_{--}^{1/2}$. By the multiplicative trace-norm bound,

$$\|\rho_{+-}\|_1 \leq \|\rho_{++}^{1/2}\|_2 \|K\|_{\infty} \|\rho_{--}^{1/2}\|_2 \leq \sqrt{\text{Tr} \rho_{++}} \sqrt{\text{Tr} \rho_{--}} = \sqrt{(1-L)L}.$$

For the pure state $|\psi\rangle = \sqrt{1-\lambda}|\psi_+\rangle + \sqrt{\lambda}|\psi_-\rangle$ with $|\psi_{\pm}\rangle \in H_{\pm}$ unit, direct computation gives $L = \lambda$ and $\|P_+|\psi\rangle\langle\psi|P_-\|_1 = \sqrt{\lambda(1-\lambda)}$. $\square$

Proof of Theorem 6

($\Leftarrow$) $L(\Phi(\rho)) = \text{Tr}(P_- \Phi(\rho)) = \text{Tr}(\Phi^{\dagger}(P_-)\rho) \leq \text{Tr}(P_- \rho) = L(\rho)$.

($\Rightarrow$) Since Φ is real CP and P_- is real symmetric, $\Phi^{\dagger}(P_-)$ is real symmetric. The inequality $\text{Tr}((\Phi^{\dagger}(P_-) - P_-)\rho) \leq 0$ for every real density ρ forces $\Phi^{\dagger}(P_-) - P_- \leq 0$: test on rank-one $\rho = |v\rangle\langle v|$ for arbitrary real unit v . $\square$

Proof of Theorem 7

($\Rightarrow$) Compressing $\Phi^{\dagger}(P_-) \leq P_-$ to H_+ :

$$P_+ \Phi^{\dagger}(P_-) P_+ = \sum_k (P_- K_k P_+)^{\top} (P_- K_k P_+) \leq P_+ P_- P_+ = 0.$$

Each summand is PSD and sums to zero, so each vanishes.

($\Leftarrow$) In block form $K_k = \begin{pmatrix} K_{++}^{+,k} & K_{+-}^{+,k} \\ 0 & K_{--}^{+,k} \end{pmatrix}$, so $K_k^{\top} P_- K_k = \begin{pmatrix} 0 & 0 \\ 0 & K_{--}^{\top,+,k} K_{--}^{+,k} \end{pmatrix}$. Trace preservation gives $\sum_k (K_{+-}^{\top,+,k} K_{+-}^{+,k} + K_{--}^{\top,+,k} K_{--}^{+,k}) = I_-$, hence $\sum_k K_{--}^{\top,+,k} K_{--}^{+,k} \leq I_-$ and $\Phi^{\dagger}(P_-) \leq P_-$. $\square$

Proof of Theorem 9 (lower bound)

Let $\sigma \in \mathcal{F}_{AB}$. Decompose $\sigma = \alpha|\psi_+\rangle\langle\psi_+| + (1-\alpha)\tau$ with $\alpha \in [0, 1]$ and τ supported in $H_+ \ominus \text{span}\{\psi_+\}$. Since $|\psi\rangle\langle\psi| - \alpha|\psi_+\rangle\langle\psi_+|$ is supported in $\text{span}\{\psi_+, \psi_-\}$ and $(1-\alpha)\tau$ is orthogonal to that span,

$$\| |\psi\rangle\langle\psi| - \sigma \|_1 = \| |\psi\rangle\langle\psi| - \alpha|\psi_+\rangle\langle\psi_+| \|_1 + (1-\alpha).$$

In the basis $\{\psi_+, \psi_-\}$,

$$|\psi\rangle\langle\psi| - \alpha|\psi_+\rangle\langle\psi_+| = \begin{pmatrix} 1-\lambda-\alpha & \sqrt{\lambda(1-\lambda)} \\ \sqrt{\lambda(1-\lambda)} & \lambda \end{pmatrix},$$

with eigenvalues $\frac{1-\alpha}{2} \pm \frac{1}{2}\sqrt{(1-\alpha)^2 + 4\alpha\lambda}$, so its trace norm equals $\sqrt{(1-\alpha)^2 + 4\alpha\lambda}$. Therefore $f(\alpha) := \frac{1}{2}[\sqrt{(1-\alpha)^2 + 4\alpha\lambda} + (1-\alpha)]$ is strictly decreasing on $[0, 1]$ for $\lambda \in (0, 1)$, minimised at $\alpha = 1$ with value $\sqrt{\lambda}$. $\square$

Proof of Theorem 13

Duhamel expansion: $e^{G_\epsilon t} = e^{G_0 t} + \epsilon \int_0^t e^{G_0(t-s)} B e^{G_0 s} ds + O(\epsilon^2)$. Since $[G_0, M] = 0$, $e^{G_0 t} \psi_0 \in H_+$, so $P_- e^{G_0 t} \psi_0 = 0$. Therefore $P_- \psi(t) = \epsilon \int_0^t P_- e^{G_0(t-s)} B e^{G_0 s} \psi_0 ds + O(\epsilon^2)$, and $L = \|P_- \psi(t)\|^2 = O(\epsilon^2)$. $\square$

C SDP soundness oracle

Theorem 20 (Moment-matrix soundness). *Let $\mathcal{W}$ be a finite set of real polynomial words in operators $\{E_a^x, F_b^y, G_c^z, J_A, J_B, J_C, \dots\}$. For any realisation with real density operator ρ , the matrix $\Gamma_{u,v} = \text{Tr}(u^\top v \rho)$, $u, v \in \mathcal{W}$, is positive semidefinite.*

Proof. For real coefficients $c = (c_u)$, $c^\top \Gamma c = \text{Tr}((\sum_u c_u u)^\top (\sum_v c_v v) \rho) = \text{Tr}(W^\top W \rho) \geq 0$ since $W^\top W \succeq 0$. $\square$

This gives a sound outer-relaxation oracle for leakage-constrained polynomial optimisation over states and effects. Convergence of the hierarchy is not claimed. Network-bilocal independence requires inflation or scalar-extension constructions (see Renou–Xu–Ligthart [17] and Ligthart–Gachechiladze–Gross [18]); extending those to carry the algebraic constraints $J_X^2 = -I$, $J_X^\top = -J_X$ together with leakage budgets is left to future work.

Figure 5 (below) verifies the $O(\epsilon^2)$ perturbative scaling of Theorem 13 numerically.

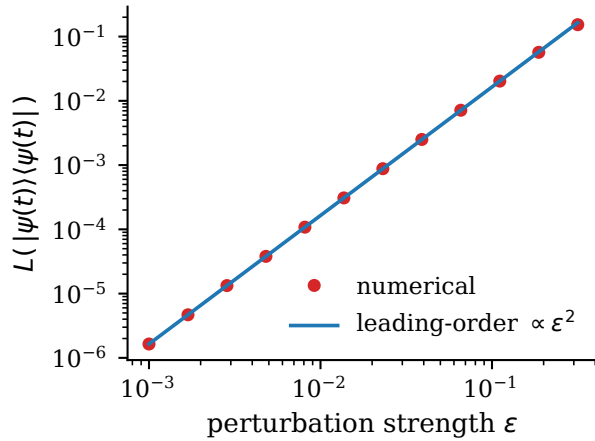


Figure 5: Leakage generated by a perturbation $G_0 + \epsilon B$ acting on an initially aligned ($L = 0$) state, with $[G_0, M] = 0$. The log-log slope is 2, confirming the Duhamel expansion of Theorem 13.